

From paired-end sequencing to variant effect prediction - a comprehensive report

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Abstract

Rare genetic disorders present a significant clinical challenge due to their extreme heterogeneity and the need for rapid and accurate molecular diagnosis. In this study, we analysed a cohort of 10 family trios, simulated from exome sequencing data, suspected of harbouring Mendelian disorders to identify the causative genomic variants. We focused our analysis on chromosome 20, exploring three primary inheritance models: autosomal recessive, autosomal inherited dominant, and autosomal de novo dominant.

The aim of this study is to bridge the gap between complex genomic raw data and a definitive clinical diagnosis, with a computational framework. Our analysis successfully identified several high-impact mutations linked to severe disorders. We developed a Python-based pipeline to aggregate all the steps of the transition from raw FASTQ files to filtered variants, to isolate the possible pathogenic ones from background noise. We utilised Bowtie2 for alignment to the GRCh38 reference genome and Freebayes for joint variant calling, finally, variants were annotated and prioritized using Ensembl's Variant Effect Predictor (VEP). All the code and data referring to this study can be found in [GitHub](#).

1. INTRODUCTION

Rare genetic disorders, though individually infrequent with a global prevalence of approximately 1 in 2500, collectively represent a significant challenge to global healthcare systems due to their vast clinical heterogeneity and diagnostic complexity. Characterized by a wide range of severity and symptoms, these conditions often necessitate a rapid and accurate molecular diagnosis to mitigate mortality and improve patient quality of life. Most of these disorders follow a Mendelian mode of inheritance—where a single gene disruption directly dictates a pathogenic phenotype—typically manifesting through autosomal dominant (AD) or autosomal recessive (AR) patterns.

In clinical genomics, Whole Exome Sequencing has emerged as a gold-standard diagnostic tool. By targeting the protein-coding regions that comprise roughly of the human genome, it provides high-depth coverage of the areas most likely to harbour disease-causing mutations. This approach is particularly efficacious when applied to family "trios" (child and both parents), as it allows for the immediate filtering of variants based on inheritance logic.

In this study, we analysed a cohort of 10 family trios

where parents are presented with a clear diagnosis, yet the child is suspected of genetic disorder. Within this framework, three inheritance models were explored:

- **Autosomal Recessive (AR):** Both parents act as asymptomatic heterozygous carriers, each transmitting a mutant allele to the child, who subsequently becomes homozygous or heterozygous for the mutation or complete disease free.
- **Autosomal Dominant (AD) Inheritance:** In this pattern, the presence of a disease phenotype requires only a single pathogenic variant in one allele. This implies that at least one parent is affected and transmits the dominant allele to the child. Typically, since we are considering rare mutations, only one parent carries the mutation.
- **Autosomal Dominant (AD) via De Novo Mutation:** As both parents are phenotypically healthy, any dominant trait in the child is investigated as a de novo mutation (DNM)—a spontaneous alteration occurring during gametogenesis or early embryonic development. These mutations are of particular interest as they bypass long-term evolutionary purifying selection, often resulting in more deleterious functional impacts than inherited variants.

Our analysis specifically targeted chromosome 20. Utilizing a Unix-based bioinformatics pipeline, we aligned sequences to the GRCh38/hg38 reference genome and utilized the Variant Effect Predictor (VEP) to assess functional impact. By integrating high-resolution genomic data with established Mendelian laws, this report investigates whether targeted bioinformatic approaches can bridge the gap between complex genetic data and definitive clinical diagnosis.

2. MATERIALS AND METHODS

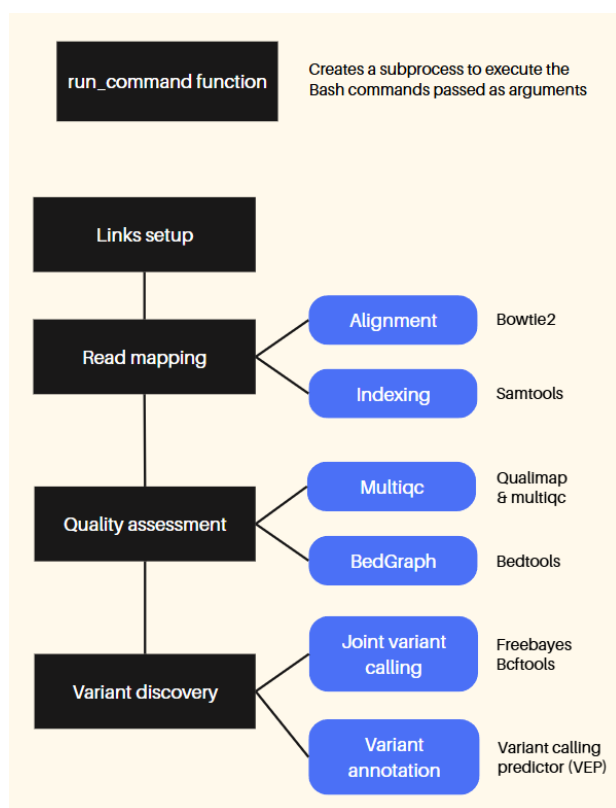


Figure 1: Comprehensive overview of the pipeline employed for the analyses.

2.1. Materials

The primary dataset for this study consisted of simulated exome sequencing data representing ten family trios (both parents and the child), in the format of three compressed FASTQ files for each case, containing raw paired-end sequencing reads for the child and both parents. In addition, the clinical status of the parents was indicated as well as the suspected disease model of inheritance. Each family trio was named accordingly to the following system: from 1 to 5 “C” cases and from 1 to 5 “M” cases, for a total of 10 trios.

Finally, the following files were provided: the GRCh38/hg38 reference genome for chromosome 20 (including its Bowtie2 indexes) against which the genomic alignment was performed, and a BED file defining the coordinates of target exons regions.

Computational processing and data integration were supported by several standardized genomic formats and tools. Data quality was monitored via FastQC⁷ and Qualimap⁸ reports, which were summarized using MultiQC, data were maintained in SAM/BAM formats. For annotation and visual inspection were employed: the Ensembl Variant Effect Predictor (VEP) by the web interface, the UCSC Genome Browser, and the Integrative Genomics Viewer (IGV). All scripts and processed data are provided below and all the processed materials referring to this study can be found in [GitHub](#).

2.2. Methods

We modularized the code into specific functions that can be called in the main block in any preferred order. First of all the user must create a new folder in a dedicated server where the process can run.

```

import subprocess

if __name__ == "__main__":
    setup_links("trio_3")
    samples = ["HG00421", "HG00422", "HG00423"]
    roles = ["child", "father", "mother"]
    run_alignment(samples, roles)
    run_qc(roles)
    run_vcf(roles)
  
```

In the main section the user specifies the name of the trio from the server and the samples ID. We used trio_3 (later referred to as 3C) as an example and we report the corresponding samples IDs.

```

def run_command(cmd):
    try:
        print(f"Running: {cmd}")
        subprocess.run(cmd, shell=True,
            check=True)
    except subprocess.CalledProcessError as e:
        print(f"ERROR: {cmd}\nDettaglio: {e}")
        exit(1)
  
```

This function is the core of our program: it uses the subprocess module to execute Bash commands passed as arguments. The "shell = True" setting allows the program to open a Bash shell to operate in the Unix environment, while "check = True" controls the exit status of the command and proceeds with the pipeline only if the previous command was successful. Errors are handled by reporting the issue to the user and by closing the process(exit(1)). This was done to guarantee the correct flow of the pipeline without any errors in the server where the code is run.

```
def setup_links(trio_id):
    run_command(f"ln -s /home/BCG2026_exam/BCG2026_Cenedese_A/{trio_id}/* . ")
    run_command(" ln -s /home/BCG2026_exam/chr20* .")
    print(f"Configurazione link per {trio_id}...")
    run_command("ls")
```

The setup-links function operates the environment configuration for the information by linking to the necessary data in the server. In more details, the function creates links with the trio paired-end sequencing, the GRCh38/hg38 reference genome for chromosome 20 (including its Bowtie2 indexes) and a BED file defining the coordinates of target exons regions.

```
def run_alignment(samples, roles):
    print("Running variant calling...")
    for i in range(0, len(samples)):
        cmd = f"bowtie2 -x chr20 -1 {samples[i]}.targets_R1.fq.gz -2 {samples[i]}.targets_R2.fq.gz --rg-id {samples[i]} --rg SM:{roles[i]} | samtools view -Sb - | samtools sort -o {roles[i]}.bam"
        run_command(cmd)

def run_qc(roles):
    print("MultiQC and Quality Control...")
    run_command("fastqc *.bam")
    for role in roles:
        cmd1 = f"qualimap bamqc -bam {role}.bam --feature-file chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed --outdir {role}_stats"
        run_command(cmd1)
        cmd2 = f"bedtools genomecov -ibam {role}.bam -bg -trackline -trackopts 'name={role}' -max 100 > {role}Cov.bg"
```

```
run_command(cmd2)
run_command("multiqc .")
```

The run_alignment and run_qc functions operate the variant calling and quality assessment analysis respectively. Bowtie2 aligns the sequence reads with the reference genome, while samtools view -Sb converts the file from SAM to BAM format. The for loop iterates these operations across the child, father and mother samples automatically. The qualimap tool performs the mapping on chromosome 20, our focus of study. At the end every multiqc provides a quality report for the DNA samples of the family trio. The bedtools genomecov command provides a bedgraph file useful to analyse the coverage track in the UCSC browser.

```
def run_vcf(roles):
    samples_str = ",".join(roles)
    print("Avvio MultiVCF...")
    run_command("nohup freebayes -f chr20.fa -m 20 -C 5 -Q 10 -q 10 --min-coverage 10 child.bam father.bam mother.bam > output.vcf ")
    run_command("bgzip output.vcf ")
    run_command("bcftools index output.vcf.gz ")
    run_command(f"bcftools view -R chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed output.vcf.gz | bcftools view -s {samples_str} | bcftools view -i 'GT[0]=\"RA\" && GT[1]=\"RR\" && GT[2]=\"RR\"' | bcftools filter -i 'QUAL>20' -Ov -o output.cand.vcf ")
```

The run-vcf function performs variant calling using Freebayes. It processes the BAM files in the order child, father and mother, comparing the samples against chromosome 20 to identify variations (SNP or indel) in the sequences. We are specifying with -m 20 to exclude low-mapping quality reads, with -min-coverage 10 to not consider sequences with less than 10 reads. -C 5 requires at least 5 observations of the alternative allele to call a variant and -q 10 is the minimum base quality score threshold. The user must update the bcftools command according to the family inheritance model being analysed in the following manner:

```

- AR --> run_command(f"bcftools view -R
chr20_ILMN_Exome_2.0_Plus_Panel.
hg38_padded.bed output.vcf.gz |
bcftools view -s {samples_str} |
bcftools view -i 'GT[0]="AA\" && GT
[1]="RA\" && GT[2]="RA\"' | bcftools
filter -i 'QUAL>20' -Ov -o output.cand.
vcf

- AD inherited, father affected -->
run_command(f"bcftools view -R
chr20_ILMN_Exome_2.0_Plus_Panel.
hg38_padded.bed output.vcf.gz |
bcftools view -s {samples_str} |
bcftools view -i 'GT[0]="RA\" && GT
[1]="RA\" && GT[2]="RR\"' | bcftools
filter -i 'QUAL>20' -Ov -o output.cand.
vcf

- AD de novo --> run_command(f"bcftools
view -R chr20_ILMN_Exome_2.0_Plus_Panel.
hg38_padded.bed output.vcf.gz |
bcftools view -s {samples_str} |
bcftools view -i 'GT[0]="RA\" && GT
[1]="RR\" && GT[2]="RR\"' | bcftools
filter -i 'QUAL>20' -Ov -o output.cand.
vcf

```

This is due to the need to specify the reference allele (R) and the alternative one (A) among the different inheritance modes between the child (GT[0]), the father (GT[1]) and mother (GT[2]). In autosomal recessive inheritance both the mother and father have a reference allele and an alternative one, the child to be affected is expected to show a pair of recessive alternative alleles (AA). In the AD inherited model, a parent must have the alternative allele and consequently show the disease symptoms. In this example we considered the father to be affected, as GT[1] shows an alternative allele. In the AD de novo inheritance mode the mutation occurred only in the child, so he is the only carrier of an alternative allele, therefore he is the only one affected by the disease.

The following output, output.cand.vcf, can be uploaded to the Ensembl Variant Effect Predictor (VEP). The used assembly is GRCh38 and the transcript database to use is the "RefSeq transcripts". In the "frequency data co-located variants" section the first three choices were selected and in the "phenotype data and citations" section the "phenotypes" check was selected.

After getting the results, they were filtered for high-impact variants and the associated phenotypes

were checked to control eventual rare diseases. The associated phenotypes needed to have low frequency according to AF and gnomAD frequencies.

Quality assessment methods

Quality assessment was performed using MultiQC analysis alongside the evaluation of coverage tracks. Our pipeline generates both MultiQC reports in HTML format and coverage tracks as .bg (bed-Graph) files. These outputs can be retrieved locally using an FTP client, such as FileZilla. Subsequently, the results are analyzed by viewing the HTML reports and uploading the .bg and .vcf files to the 'Custom Tracks' section of the UCSC Genome Browser. This integration of coverage and variant data allows for a comprehensive quality control and visual validation of the results.

3. RESULTS

3.1. Quality assessment MultiQC results

To assess whether future diagnosis could be reliable, detailed quality and coverage analysis were conducted, case 1M was selected as a representative example of all 10 trios. Results are shown in Fig. 2 – 5.

FastQC Sequence quality:

Initial quality control was performed to evaluate mapping efficiency after the bowtie-2 alignment. The Per Sequence Quality Score distribution (Fig. 4) indicates that most reads maintain a high base-calling accuracy, indicated by a PHRED score above 30 (less than 0,001 probability of error in base calling). This is further supported by the Mean Quality Score per Position (Fig. 3), which consistently remains within the optimal green zone over Phred score of 30, across the entire read length.

Qualimap Coverage Distribution:

After the alignment a coverage distribution analysis was performed on the BAM files, to validate the actual depth of usable data before proceeding to variant calling. Each member of the family trio has at least 95% of the genome covered at 30X and 99.95% covered at 20X (Fig. 5). This high breadth of coverage ensures that virtually the entire genome is sequenced with enough redundancy to detect heterozygous variants with high sensitivity.

Sample Name	% GC	Ins. size	≥ 30X	Median cov	Mean cov	% Aligned
child1	52%	300	95.1%	40.0X	40.0X	100.0%
father1	52%	299	94.9%	40.0X	40.0X	100.0%
mother1	52%	300	95.1%	40.0X	40.0X	100.0%

Figure 2: MultiQC General Statistics.

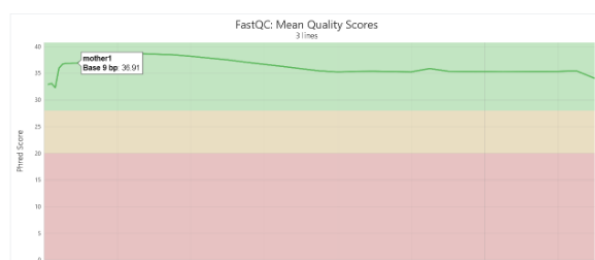


Figure 3: FastQC Mean Quality Score per Position.

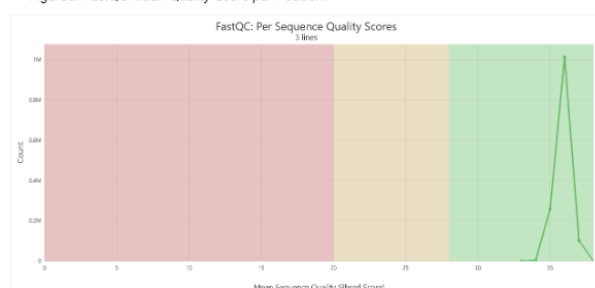


Figure 4: FastQC Phred Quality Score per Sequence.

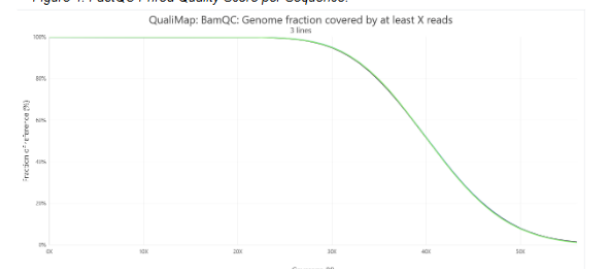


Figure 5: Qualimap Genome Fraction Covered by at Least X Reads.

MultiQC General Statistics:

After the mapping, data from FastQC and Qualimap were aggregated using MultiQC to provide a summary of the alignment success. The General Statistics (Fig.2) confirms robust data reliability:

- **Alignment Rate:** A high fraction of the reads was successfully mapped to the reference genome, indicating a good sequencing output and minimal contamination.
- **Mean Coverage:** Each member of the family trio exhibited a mean coverage depth exceeding 30X. This ensures that there are enough independent reads at each genomic position to make accurate variant calls.

3.2. Variant calling results

Table 1 shows a comprehensive overview of the variant calling results for each child of family trios. Two cases, 1C and 1M, were found to be healthy.

Developmental and epileptic encephalopathy

Case 2C is affected by developmental and epileptic encephalopathy. The inheritance patterns involve an affected dominant allele inherited from the father (autosomal dominant), who would also manifest the same pathology. Case 3M is also affected by the disease, but the inheritance mode follows a dominant frameshift de novo mutation. In this case, the mutation has not been inherited by a parent, but has occurred spontaneously in the child genome, in particular in the KCNQ2 gene. In the first case, it is caused by a frameshift variant resulting from the insertion of an 'A' in the GAG sequence that becomes GAAG at position 20:46029385-46029387 in the SLC12A5 gene. In the second one, the affected gene is different and the de novo mutation happened with the deletion of a G in the sequence CGGCC. This disease causes impairment of cognitive functions influenced by both seizure and interictal epileptiform activity and the neurobiological process behind the epilepsy¹. Both the KCNQ2 and SCN1A genes encode for ion channel proteins, known to be linked to epileptiform activity.

Aicardi Goutieres syndrome

Aicardi-Goutières syndrome (AGS) is a progressive genetic encephalopathy caused by pathogenic mutations in genes controlling cellular anti-viral responses and nucleic acid metabolism. The mutations initiate autoinflammatory processes in the brain². Case 3C is affected by this disease carrying a de novo dominant mutation in the gene SAMHD1. The mutation affects a splice acceptor site, leading to aberrant splicing that triggers brain inflammation.

Glutathione synthetase deficiency with 5-oxoprolinuria

5-Oxoprolinuria (pyroglutamic aciduria) resulting from glutathione synthetase (GSS) deficiency is an

Tabella 1: Overview of the Variant Prioritization outcomes

Case	Gene	Variant location	Consequence	REF/ALT	Impact	Associated phenotype
1C (AR)	Healthy					
2C (AD father)	SLC12A5	20:46029385-87	Frameshift	GAG/GAAG	High	Developmental and epileptic encephalopathy 34
3C (AD)	SAMHD1	20:36897960	Splice acceptor	C/A	High	Aicardi-Goutieres syndrome 5
4C (AR)	GSS	20:34931401-05	Frameshift	TGGTC/TGTC	High	Glutathione synthetase deficiency
5C (AR)	STK4	20:44981926	Stop gained	C/T	High	Combined immunodeficiency (STK4)
1M (AR)	Healthy					
2M (AD mother)	PANK2	20:3889654-3889656	Frameshift	ATC/AC	High	Healthy carrier of pantothenate kinase associated neurodegeneration
3M (AD)	KCNQ2	20:63407144-48	Frameshift	CGGCC/CGCC	High	Developmental and epileptic encephalopathy 34
4M (AR)	MKKS	20:10407706	Frameshift	CAG/CG	High	Bardet-Biedl / McKusick-Kaufman syndrome
5M (AR)	MKKS	20:10407692	Frameshift	A/C	High	Bardet-Biedl / McKusick-Kaufman syndrome

inherited autosomal recessive disorder characterized by massive urinary excretion of 5-oxoproline, metabolic acidosis, haemolytic anaemia and central nervous system damage³. Case 4C is affected by this disease due to a deletion in the GSS gene that causes a frameshift mutation. In this family trio the inheritance mode is autosomal recessive, so both parents must carry at least a recessive allele.

Combined immunodeficiency due to STK4 deficiency

Case 5C is affected by combined T and B cell immunodeficiency. It presents with recurrent opportunistic viral, bacterial and fungal infections involving skin, upper and lower respiratory tract or septicemia⁴. The mutation in the 5C case brings to the presence of an aberrant stop signal in the gene transcription into RNA. In particular, the mutation involves a C that is replaced by a T. The inheritance mode is autosomal recessive, so the child is symptomatic due to be homozygous for the risk allele.

Bardet-Biedl syndrome / McKusick Kaufman syndrome / MKKS-related disorder

McKusick-Kaufman syndrome is a syndrome inherited in an autosomal recessive pattern caused by mutations in the MKKS gene mapped onto chromosome 20p12. Mutations in the same gene causes Bardet-Biedl-6 syndrome (BBS-6) inherited in an autosomal recessive pattern. BBS-6 comprises the phenotypic triad hydrometrocolpos, postaxial polydactyly and congenital cardiac disease, that can also occur in McKusick Kaufman syndrome, there is a significant clinical overlap the diseases in childhood. Borderline cases of MKS and BBS have been reported⁵. Case 4M and 5M both show a mutation

in the MKKS gene that can cause these diseases, or borderline cases. Although the mutations are located close to each other on the chromosome 20 locus, their impact on the protein function differs. Case 4M shows a frameshift deletion of an A in the CAG sequence; case 5M a nonsense mutation (stop gained) by a substitution of an A into a C. The inheritance mode respects the autosomal recessive nature of these diseases.

Pantothenate kinase associated neurodegeneration

Pantothenate Kinase-Associated Neurodegeneration (PKAN) is an autosomal recessive disorder characterized by a mutation in the PANK2 gene. The clinical presentation may range from only speech disorder to severe generalized dystonia, spasticity, visual loss, dysphagia and dementia⁶. Even if we aspected a de novo autosomal dominant case, we found an autosomal recessive condition. The child is asymptomatic but carries the recessive allele that can cause the syndrome symptoms in a homozygotic recessive case. The child must have inherited the mutated allele by his mother who is affected by the disease, therefore she must carry both the recessive alleles.

3.3. UCSC Genome Browser results

Following the variant calling process, a targeted coverage analysis was conducted on the mutated regions to validate the calls. The coverage tracks for case 3M, which harbors a detected de novo frameshift variant, are presented in Fig. 6 as a representative example. Around the specific deletion site, the coverage exceeded 40X for all trio members, providing the necessary depth to confirm the accuracy and biological validity of the variant.

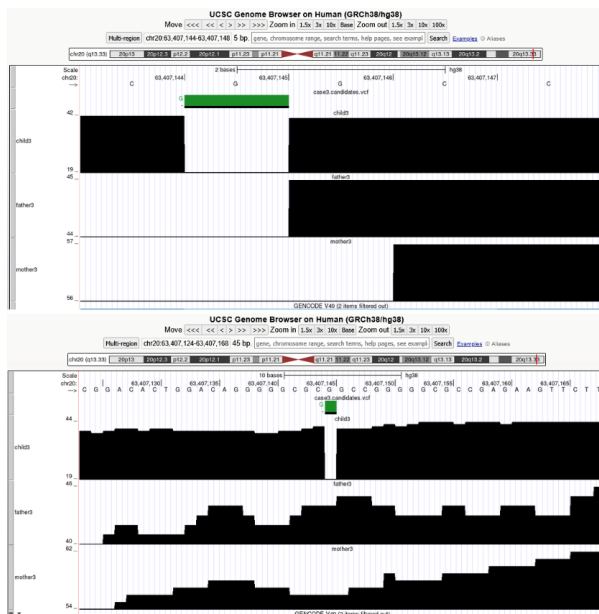


Figure 6: coverage tracks at deletion position.

REFERENCES

All the code and data referring to this study can be found in [GitHub](#).

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